EXPERIMENT 5: DEVELOP A CUSTOMIZED PING COMMAND

AIM:

To develop a Python program that checks server connectivity and measures response time.

PROCEDURE:

1. Install Python and save the program as custom_ping.py.
2. Enter a hostname or IP address when prompted.
3. The program sends a small test request using the system ping utility.
4. It displays the command output and reports whether the host responded.
5. Run the program only against hosts you are allowed to test.

6. Test the program first with the loopback address to confirm that the local networking stack responds.
7. Repeat the test with an authorized server hostname and compare the response status.
8. Note that some servers or firewalls block ICMP, so a failed ping does not always mean that the server is offline.

CODE:

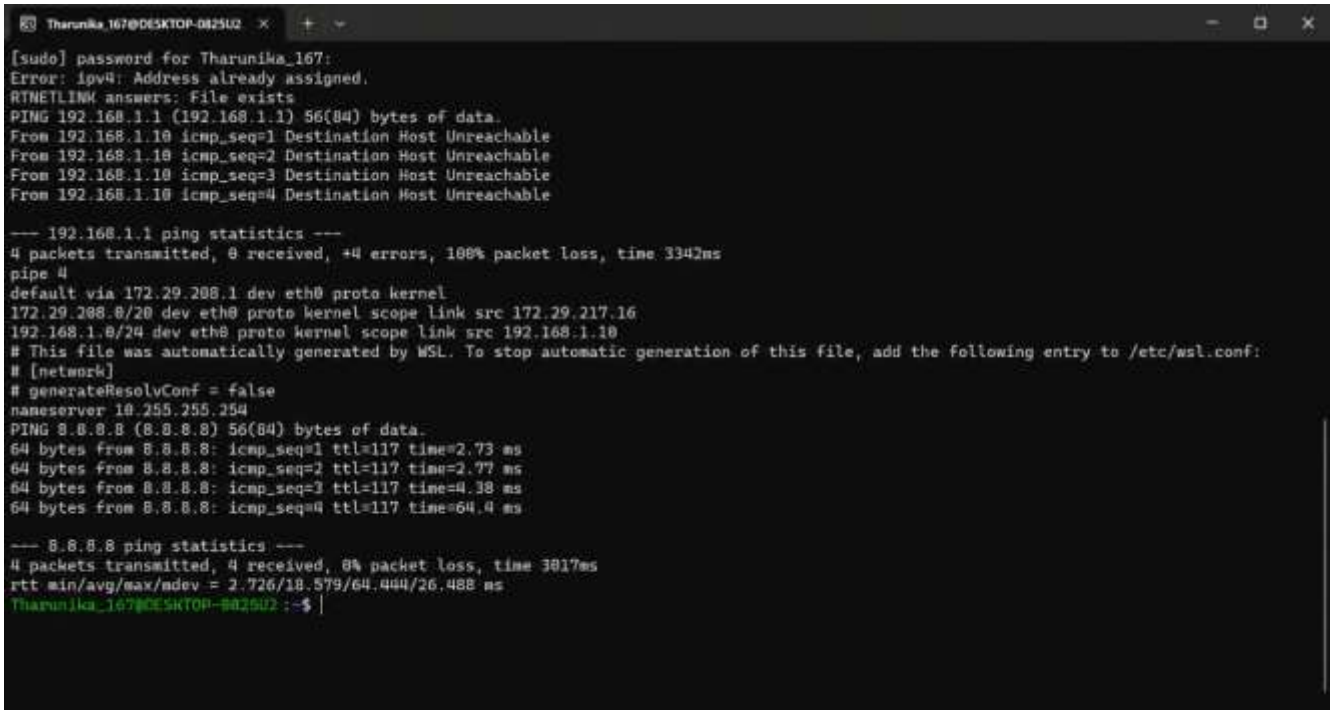
```
import platform
import subprocess

host = input("Enter hostname or IP address: ").strip()
if not host:
    raise SystemExit("Host cannot be empty.")

count_option = "-n" if platform.system().lower() ==
"windows" else "-c"
command = ["ping", count_option, "4", host]
```

```
try:
    result = subprocess.run(command,
capture_output=True, text=True, timeout=20)
    print(result.stdout or result.stderr)
    if result.returncode == 0:
        print("Status: Host responded to ping.")
    else:
        print("Status: No successful ping response.")
except subprocess.TimeoutExpired:
    print("Status: Request timed out.")
```

OUTPUT:



RESULT:

Thus, a customized Python program was developed to test host connectivity using ping.